

277. $w(x, y) = ye^x, \theta = \frac{\pi}{3}$ 278. $f(x, y) = x \arctan(y), \theta = \frac{\pi}{2}$ 279. $f(x, y) = \ln(x + 2y), \theta = \frac{\pi}{3}$

For the following exercises, find the gradient.

280. Find the gradient of $f(x, y) = \frac{14 - x^2 - y^2}{3}$.
Then, find the gradient at point $P(1, 2)$.
281. Find the gradient of $f(x, y, z) = xy + yz + xz$ at point $P(1, 2, 3)$.
282. Find the gradient of $f(x, y, z)$ at P and the directional derivative in the direction of $\mathbf{u}$:
 $f(x, y, z) = \ln(x^2 + 2y^2 + 3z^2), P(2, 1, 4), \mathbf{u} = \frac{-3}{13}\mathbf{i} - \frac{4}{13}\mathbf{j} - \frac{12}{13}\mathbf{k}$.
283. $f(x, y, z) = 4x^5y^2z^3, P(2, -1, 1), \mathbf{u} = \frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$

For the following exercises, find the directional derivative of the function at point P in the direction of $\mathbf{Q}$.

284. $f(x, y) = x^2 + 3y^2, P(1, 1), Q(4, 5)$ 285. $f(x, y, z) = \frac{y}{x+z}, P(2, 1, -1), Q(-1, 2, 0)$

For the following exercises, find the derivative of the function at P in the direction of $\mathbf{u}$.

286. $f(x, y) = -7x + 2y, P(2, -4), \mathbf{u} = 4\mathbf{i} - 3\mathbf{j}$ 287. $f(x, y) = \ln(5x + 4y), P(3, 9), \mathbf{u} = 6\mathbf{i} + 8\mathbf{j}$
288. [T] Use technology to sketch the level curve of $f(x, y) = 4x - 2y + 3$ that passes through $P(1, 2)$ and draw the gradient vector at P .
289. [T] Use technology to sketch the level curve of $f(x, y) = x^2 + 4y^2$ that passes through $P(-2, 0)$ and draw the gradient vector at P .

For the following exercises, find the gradient vector at the indicated point.

290. $f(x, y) = xy^2 - yx^2, P(-1, 1)$ 291. $f(x, y) = xe^y - \ln(x), P(-3, 0)$

292. $f(x, y, z) = xy - \ln(z), P(2, -2, 2)$ 293. $f(x, y, z) = x\sqrt{y^2 + z^2}, P(-2, -1, -1)$

For the following exercises, find the derivative of the function.

294. $f(x, y) = x^2 + xy + y^2$ at point $(-5, -4)$ in the direction the function increases most rapidly
295. $f(x, y) = e^{xy}$ at point $(6, 7)$ in the direction the function increases most rapidly
296. $f(x, y) = \arctan\left(\frac{y}{x}\right)$ at point $(-9, 9)$ in the direction the function increases most rapidly
297. $f(x, y, z) = \ln(xy + yz + zx)$ at point $(-9, -18, -27)$ in the direction the function increases most rapidly
298. $f(x, y, z) = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$ at point $(5, -5, 5)$ in the direction the function increases most rapidly

For the following exercises, find the maximum rate of change of f at the given point and the direction in which it occurs.

299. $f(x, y) = xe^{-y}, (1, 0)$ 300. $f(x, y) = \sqrt{x^2 + 2y}, (4, 10)$ 301. $f(x, y) = \cos(3x + 2y), \left(\frac{\pi}{6}, -\frac{\pi}{8}\right)$

For the following exercises, find equations of

- the tangent plane and
- the normal line to the given surface at the given point.

- 302.** The level surface $f(x, y, z) = 12$ for $f(x, y, z) = 4x^2 - 2y^2 + z^2$ at point $(2, 2, 2)$.
- 303.** $f(x, y, z) = xy + yz + xz = 3$ at point $(1, 1, 1)$
- 304.** $f(x, y, z) = xyz = 6$ at point $(1, 2, 3)$
- 305.** $f(x, y, z) = xe^y \cos z - z = 1$ at point $(1, 0, 0)$

For the following exercises, solve the problem.

- 306.** The temperature T in a metal sphere is inversely proportional to the distance from the center of the sphere (the origin: $(0, 0, 0)$). The temperature at point $(1, 2, 2)$ is 120°C .
- Find the rate of change of the temperature at point $(1, 2, 2)$ in the direction toward point $(2, 1, 3)$.
 - Show that, at any point in the sphere, the direction of greatest increase in temperature is given by a vector that points toward the origin.
- 307.** The electrical potential (voltage) in a certain region of space is given by the function $V(x, y, z) = 5x^2 - 3xy + xyz$.
- Find the rate of change of the voltage at point $(3, 4, 5)$ in the direction of the vector $\langle 1, 1, -1 \rangle$.
 - In which direction does the voltage change most rapidly at point $(3, 4, 5)$?
 - What is the maximum rate of change of the voltage at point $(3, 4, 5)$?
- 308.** If the electric potential at a point (x, y) in the xy -plane is $V(x, y) = e^{-2x} \cos(2y)$, then the electric intensity vector at (x, y) is $\mathbf{E} = -\nabla V(x, y)$.
- Find the electric intensity vector at $(\frac{\pi}{4}, 0)$.
 - Show that, at each point in the plane, the electric potential decreases most rapidly in the direction of the vector $\mathbf{E}$.
- 309.** In two dimensions, the motion of an ideal fluid is governed by a velocity potential φ . The velocity components of the fluid u in the x -direction and v in the y -direction, are given by $\langle u, v \rangle = \nabla \varphi$. Find the velocity components associated with the velocity potential $\varphi(x, y) = \sin \pi x \sin 2\pi y$.

4.7 Maxima/Minima Problems

Learning Objectives

- 4.7.1** Use partial derivatives to locate critical points for a function of two variables.
- 4.7.2** Apply a second derivative test to identify a critical point as a local maximum, local minimum, or saddle point for a function of two variables.
- 4.7.3** Examine critical points and boundary points to find absolute maximum and minimum values for a function of two variables.

One of the most useful applications for derivatives of a function of one variable is the determination of maximum and/or minimum values. This application is also important for functions of two or more variables, but as we have seen in earlier sections of this chapter, the introduction of more independent variables leads to more possible outcomes for the calculations. The main ideas of finding critical points and using derivative tests are still valid, but new wrinkles appear when assessing the results.